

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Friday 10 January 2025

Morning (Time: 2 hours)

Paper reference **31760H**

Information Technology
UNIT 1: Information Technology Systems

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P81197A

©2025 Pearson Education Ltd.
1/1/1



Answer ALL questions. Write your answers in the spaces provided.

1 Designs for Us is a company that creates logos and advertising materials.

(a) State **two** input devices the company's designers could use to create logos.

(2)

1

.....

.....

2

.....

.....

(b) (i) The designers store files on a central server.

Explain **two** benefits to the designers of storing files on the central server.

(4)

1

.....

.....

.....

.....

.....

2

.....

.....

.....

.....

.....

.....



(ii) Explain **two** benefits to the company of storing files on the central server.

(4)

1

2



P 8 1 1 9 7 A 0 3 2 4

(c) The designers mainly work remotely. They use a virtual private network (VPN) to connect to the company network.

Explain **two** benefits to the designers of using a VPN.

(4)

1

2

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



(d) The designers save their images using different file formats.

Discuss the features of **two** file formats the designers could use to save their image files.

(8)

(Total for Question 1 = 22 marks)



P 8 1 1 9 7 A 0 5 2 4

- 2 A sixth form college has a local area network (LAN) that students use to access resources and learning materials.

The students save their work to the network when in college or at home.

- (a) Utility software is installed on the network.
(i) Anti-virus software is one type of utility software.

Explain the purpose of **two other** types of utility software.

(4)

1

2

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



(ii) Describe how the anti-virus software works to reduce the risk of viruses infecting the college network.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



- (b) Staff enter the server room using a keypad entry system. The four-digit code is changed each week.

The college is updating to an ID card entry system.

Explain **two** ways an ID card entry system makes access to the server room more secure than a keypad entry system.

(4)

1

.....

.....

.....

.....

.....

2

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



(c) Students are required to log into the college network.

Describe the login authentication process.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



P 8 1 1 9 7 A 0 9 2 4

(d) Discuss the moral and ethical factors for staff and students when using IT in the classroom.

You need to consider:

- privacy
- online behaviour and netiquette
- acceptable use.

(8)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 2 = 24 marks)



P 8 1 1 9 7 A 0 1 1 2 4

- 3 CG is a company that sells electrical products. It has a chain of stores and a warehouse.

Each store has a stockroom with a desktop PC and printer.

Electronic Point of Sale (EPOS) terminals are used by sales staff to complete sales and process customer payments.

The EPOS system is connected to the desktop PC in the stockroom. When a sale is made stock levels are updated automatically on the desktop PC.

At the end of each day, a stock request is sent to the database server in the warehouse from the desktop PC in the stockroom. Stock requests are printed from a desktop PC in the company warehouse.

- (a) Draw a diagram to show how a store and the warehouse interact.

The diagram should include devices and systems that are used, including:

- desktop PCs
- EPOS terminals
- payment card readers
- printers
- database server
- devices to be connected and the connection type(s) used
- annotations.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Use this space to draw your diagram for Q3a.

(6)





DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- (c) CG has a secure website that uses a digital certificate where customers can place online orders.

Explain **two** reasons why websites use digital certificates.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



(d) Discuss the security protocols websites use for data transmission.

(8)

Handwriting practice area with horizontal dotted lines.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 3 = 22 marks)



P 8 1 1 9 7 A 0 1 7 2 4

4 Clothes for Us is a company that designs clothes for high-street shops.

The company is considering moving its data to a cloud-based storage system.

- (a) Discuss the positive and negative impacts to the company and staff of moving to cloud-based storage.

(10)

Area for writing the answer to question 4(a). The area contains horizontal dotted lines for writing.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Handwriting practice lines consisting of eight sets of three horizontal dotted lines.



P 8 1 1 9 7 A 0 1 9 2 4

(b) Clothes for Us has a company network. The company’s designers use the network to work collaboratively.

Evaluate the features of the company network that will allow the designers to work collaboratively.

(12)

Handwriting practice area with 20 horizontal dotted lines.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Handwriting practice area with 20 horizontal dotted lines.

(Total for Question 4 = 22 marks)

TOTAL FOR PAPER = 90 MARKS





DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

BLANK PAGE



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

BLANK PAGE



P 8 1 1 9 7 A 0 2 3 2 4

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

BLANK PAGE

